TOURSAFE

Smart Real Time Tourist Safety Monitoring & Emergency Response System using AI, Geo-Fencing, and Blockchain-based Digital ID

LITERATURE REVIEW

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PROJECT ID	: SEC24CSSO010
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LITERATURE REVIEW

PAPER NAME	FINDINGS	DRAWBACKS
1) Striim Labs & Malhotra et al. (2016/2023–2026) Seq2Seq LSTM Autoencoders on multivariate time-series accelerometer / sensor telemetry.	Sequence-to-sequence reconstruction loss (MSE) on unlabeled 50Hz sensor streams isolates structural anomalies with a 95.3% detection rate. **TourSafe Resolution: Normalizes 3-axis readings into an orientation-invariant scalar magnitude (A_mag), compiles to ONNX Runtime, and uses a two-stage consecutive window breach protocol.**	High computational overhead on mobile CPU cores; vulnerable to single-point vibration noise spikes causing false positives.
2) Pang et al. / IEEE & MDPI (2023–2025) Spatial-temporal trajectory deviation using Hausdorff Distance Metrics & VAE Encoders.	Proves that evaluating maximum spatial displacement across time-ordered trajectories yields superior precision (0.960) and recall (0.965) for moving objects compared to static radius checks. **TourSafe Resolution: Implements a decoupled edge interceptor: client-side Turf.js handles offline GeoJSON hazard checks, while SNN-CAD+ runs asynchronously on the backend.**	Assumes continuous cloud-side server connectivity; completely breaks down in zero-signal cellular dead zones.
3) SSISH & Healthcare BC-SSI Studies (2022–2026) W3C Decentralized Identifiers (DIDs) & Verifiable Credentials (VCs) on permissioned/PoS ledgers.	Eliminates centralized database honeypots by keeping sensitive PII/medical records encrypted in off-chain IPFS storage while anchoring immutable DID hashes on-chain. **TourSafe Resolution: Introduces Smart Contract Emergency Access Grants: verified AI anomaly flags automatically issue time-bound responder QR decryption privileges.**	Lacks an automated 'break-glass' emergency access delegation mechanism when a patient is unconscious or non-verbal.
4) BNSS Sec. 173(1)(ii) / NCRB ICJS (2023–2025 Standards) Electronic First Information Report (e-FIR) legal framework & zero-jurisdiction dispatch.	Establishes full legal validity for zero-jurisdiction e-FIR generation and automated inter-agency emergency data sharing across law enforcement and EMS nodes. **TourSafe Resolution: Automates e-FIR compilation via a Node.js**	Relies on manual web portal entry by human victims or dispatchers, introducing a 20-30 minute administrative lag.

	microservice paired with a Haversine Distance Router to dispatch JSON/PDF payloads in < 60 seconds.	
5) US Patent US11842621B2 (2023) Multi-stage G-force thresholding combined with temporal IMU feature classification.	Validates that combining sharp impact G-force acceleration spikes with post-impact stillness checks significantly reduces false alarm rates. **TourSafe Resolution: Open-architecture integration pairing edge IMU 50Hz feature streams with Polygon PoS W3C DIDs and B2G command dashboards.**	Proprietary closed-loop hardware ecosystems; lacks decentralized identity resolution or cross-border B2G integration.
6) Yuejun Guo & Anton Bardera (SHNN-CAD+, Sensors 2019) Conformal prediction dealing with online trajectory anomaly detection.	Introduces conformal p-value thresholding, a re-do strategy to dynamically reclassify false positives, and adaptive anomaly thresholding. **TourSafe Resolution: Implements Directed Hausdorff Distance with constraint window DHD(omega) to lower complexity to O((m+n)omega).**	Standard DHD ignores trajectory direction and suffers from a high computational footprint O(mn).
7) Laxhammar & Falkman (SNN-CAD, 2011/2014) Sequential Conformal Anomaly Detection in trajectories using Hausdorff distance.	Standardizes unsupervised anomaly detection on raw unequal-length trajectories without pre-modeling. **TourSafe Resolution: Formulates an adaptive data-adaptive threshold (epsilon_l) calculated from the minimum p-value of known normal trajectories.**	Relies on manually selected, static anomaly thresholds (epsilon), leading to poor usability across domains.
8) Inayatulloh et al. (IEEE, 2024) Blockchain ledger setup for travel agent disintermediation and partner trust.	Registers travelers and providers as immutable blocks on a ledger, allowing regulator oversight without transactional access. **TourSafe Resolution: Repurposes peer-to-peer validation loops to secure traveler safety identities.**	Highly transactional business-oriented focus; fails to provide real-time kinetic emergency response workflows.

9) Giorgia Farris et al. (IEEE, 2021) Blockchain, IoT, and ETIS indicators for sustainable tourism.	Decentralizes tourism interactions, reducing middleman costs while monitoring environmental and transport indicators. **TourSafe Resolution: Combines the IoT-blockchain framework with active wearable sensor tracking (ESP32 / MPU6050) to protect tourist lives.**	Focuses strictly on direct booking, sustainability scoring, and ledger-based payments.
10) Jaesung Choi & Pilwon Kim (AIMS Math, 2022) Early warning indicator (EWI) using machine-based predictability Pred(t).	Quantifies system state predictability using ARIMA and Echo State Networks (ESNs) to detect critical transitions before collapse. **TourSafe Resolution: Validates the core thesis of monitoring state-change predictability asynchronously on the backend (via SNN-CAD+) to detect critical transitions before collapse.**	Restricted to low-dimensional statistical time-series; fails to address multi-agent spatial trajectories.
11) Guillaume Falmagne et al. (PNAS, 2026) Interpretable early warning indicators using machine learning models.	Assesses predictability and variance under high-uncertainty environments using online user data. **TourSafe Resolution: Couples ML trajectory trend deviations with automated zero-jurisdiction e-FIR generation and dispatch routing.**	Academic, offline sandbox evaluation; lacks integration with legal, administrative, and real-time dispatch systems.
12) Mathias Marconi et al. (Chaos, 2024) Physical evaluation of critical slowing down in real experimental systems.	Proves that complex systems near critical boundaries slow down when recovering from small environmental perturbations. **TourSafe Resolution: Reconstructs physical slowing down (prolonged inactivity or immobility after high impact) as an anomaly check via the Seq2Seq LSTM decoder.**	Purely physical, hardware-centric simulation; does not translate slowing down parameters into automated edge telemetry.
13) Sandip V. George et al. (Physica Scripta, 2023) Multi-metric early warning signals for critical transitions in complex systems.	Combines variance, autocorrelation, and entropy to build high-accuracy warning signals. **TourSafe Resolution: Decouples feature calculation: simple orientation-invariant G-force metrics on edge, heavy multidimensional calculations in the**	High computational overhead makes real-time edge execution on mobile processors impractical.

	backend FastAPI gateway.	
14) A. Presno Velez et al. (AIMS Math, 2025) Vibration monitoring and LSTM networks for data-driven structural transition detection.	Leverages LSTM networks to model structural health transitions under noisy vibrations. **TourSafe Resolution: Adapts high-frequency vibration modeling to isolate 50Hz human kinetic trail anomalies from background off-road vehicular noises.**	Limited to mechanical/structural engineering systems; lacks spatial-temporal geographical mapping capabilities.
15) Alexander Musaev et al. (AIMS Math, 2024) Adaptive algorithms for change point detection in financial and physical time series.	Tracks regime shifts in noisy time series through dynamic statistical modeling. **TourSafe Resolution: Adds a client-side two-stage consecutive window check to verify change-points before triggering emergency alerts.**	Highly vulnerable to false alarms when subjected to transient, non-catastrophic human movements (e.g., jogging or dropping a phone).
16) Keogh et al. (HOT SAX, 2005) Time series discord discovery using symbolic aggregate approximation (SAX).	Brute-force searching of the most unusual subsequences in massive, unlabeled temporal datasets. **TourSafe Resolution: Replaces symbolic aggregate checks with continuous Seq2Seq LSTM Autoencoder bottlenecks, achieving sub-second inference in Kafka-Spark pipelines.**	Severe computational bottleneck; high time-complexity of SAX calculations on continuous real-time streams.
17) US Patent US11800314B2 (2023) Interlinked geofencing system mapping user locations and success data.	Generates geo-fences to automatically authenticate users or authorize transaction events. **TourSafe Resolution: Adapts interlinked geofencing to map sequential wilderness trails and restricted hazard zones dynamically using Turf.js.**	Designed primarily for commerce and static event authorization; lacks continuous kinetic anomaly checks.

18) WO2018126029A2 (Google Patents / WIPO, 2018) Ledger-based communication and secure execution setups for IoT devices.	Handles discovery, trust, policy, keys, and failover via an on-chain Root of Trust protocol. **TourSafe Resolution: Limits on-chain transactions to identity roots (W3C DIDs) and temporary emergency grants, offloading raw sensor telemetry to Redis and WebSockets.**	High transaction fee overhead and latency; poorly optimized for high-frequency 50Hz sensor streams.
19) European Patent EP4123592A1 (2023) Decentralized identity registers linked to off-chain encrypted medical databases.	Focuses on using decentralized registers to secure and manage emergency healthcare access pathways. **TourSafe Resolution: Integrates AI-driven dynamic access grants that monitor live telemetry and trigger on-chain decryption privileges autonomously.**	Lacks real-time autonomous trigger mechanisms; patient's active key interaction is required to release records.
20) Indian Patent App 202341058912A (2023) GPS-based geofencing and centralized authority dashboard visualization.	Real-time tourist safety monitoring and automated authority dispatch. **TourSafe Resolution: Implements client-side offline-first AES-256 local SQLite queuing to preserve tracking, syncing automatically upon reconnection.**	Fails to preserve tracking and telemetry during cellular signal drops in remote zones.